INCH-POUND

MIL-M-38510/341E
AMENDMENT 4
20 September 1999
SUPERSEDING
AMENDMENT 3
14 January 1994

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

Inactive for new design after 18 April 1997

This amendment forms a part of MIL-M-38510/341E, dated 28 April 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

PAGE 3

Setup time J, K, or D low to clock pulse, device type 02: Delete and substitute as follows:

"Device type 02 ---------- 4.0 ns minimum"

PAGE 6

TABLE I, I_{IL1}, minimum limits column: Delete "-0.3" and substitute "0.0".

TABLE I, I_{IL2}, minimum limits column: Delete "-0.9" and substitute "0.0".

TABLE I, I_{IL3}, minimum limits column: Delete "-.12" and substitute "0.0".

PAGE 45

TABLE III, device type 01, footnote 8/: Add the following: "f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein."

PAGE 51

TABLE III, device type 03, test number 91, J_1 and J_2 columns: Delete " " and substitute "A".

TABLE III, device type 03, test number 92, J_1 and J_2 columns: Delete " " and substitute "A".

PAGE 56

TABLE III, device type 04, footnote 7/: Add the following: "f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein."

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PAGE 63

TABLE III, device type 05, footnote 5/: Add the following: "f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein."

PAGE 66

* TABLE III, device type 06, subgroup 7: delete all rows for subgroup 7 (tests number 90 through 102).

* TABLE III, device type 06, subgroup 8: delete entire row.

* Insert new page 66A after page 66. (This is a new page and not a replacement).

PAGE 67

* TABLE III, device type 06, subgroup 7: delete all rows for subgroup 7 (tests number 90 through 102).

* TABLE III, device type 06, subgroup 8: delete entire row.

* Insert new page 67A after page 67. (This is a new page and not a replacement).

PAGE 71

TABLE III, device type 06, footnote 5/: Add the following: "f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in Table III, herein."

PAGE 72

TABLE III, device type 07, V_{OL} test: add " " for test number 21, for pins D_0, D_1, D_2, D_3, D_4, and D_5.

PAGE 74

TABLE III, device type 07, footnote 7/: Add the following: "f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to thelimits specified in table III, herein."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH, TD
Army - AR, MI, SM
Air Force - 19, 99

(Project 5962-1861)

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TABLE III. Group A inspection for device type 06 – Continued
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883	Cases 2,R,S,X	1	2	3	4	5	6	7	8	9	10
		method	Test no.	$\overline{OE}$	$\overline{Q_0}$	D_0	D_1	$\overline{Q_1}$	$\overline{Q_2}$	D_2	D_3	$\overline{Q_3}$	GND
7	Functional	3014	90.1	B	X	A	A	X	X	B	A	X	GND
T_C = +25°c	3/		90.2	"	L	A	A	L	H	B	A	L	"
			90.3	"	L	A	B	L	H	B	A	L	"
			90.4	"	L	A	B	H	H	B	A	L	"
			90.5	"	L	A	B	H	H	A	A	L	"
			90.6	"	L	A	B	H	L	A	A	L	"
			90.7	"	L	A	B	H	L	A	B	L	"
			90.8	"	L	A	B	H	L	A	B	H	"
			90.9	"	L	A	B	H	L	A	B	H	"
			90.10	"	L	A	B	H	L	A	B	H	"
			90.11	"	L	A	B	H	L	A	B	H	"
			90.12	"	L	A	B	H	L	A	B	H	"
			90.13	"	L	A	B	H	L	A	B	H	"
			90.14	"	L	A	B	H	L	A	B	H	"
			90.15	"	L	A	B	H	L	A	B	H	"
			90.16	"	L	A	B	H	L	A	B	H	"
			90.17	"	L	B	A	H	L	A	B	H	"
			90.18	"	H	B	A	L	L	A	B	H	"
			90.19	"	H	B	A	L	L	A	B	H	"
			90.20	"	H	B	A	L	L	A	B	H	"
			90.21	"	H	B	A	L	L	B	B	H	"
			90.22	"	H	B	A	L	H	B	B	H	"
			90.23	"	H	B	A	L	H	B	A	H	"
			90.24	"	H	B	A	L	H	B	A	L	"
			90.25	"	H	B	A	L	H	B	A	L	"
			90.26	"	H	B	A	L	H	B	A	L	"
			90.27	"	H	B	A	L	H	B	A	L	"
			90.28	"	H	B	A	L	H	B	A	L	"
			90.29	"	H	B	A	L	H	B	A	L	"
			90.30	"	H	B	A	L	H	B	A	L	"
			90.31	"	H	B	A	L	H	B	A	L	"
			90.32	"	H	B	A	L	H	B	A	L	"
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.												

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 – Continued
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S,X Test no.	11 CP	12 $\overline{Q_4}$	13 D_4	14 D_5	15 $\overline{Q_5}$	16 $\overline{Q_6}$	17 D_6	18 D_7	19 $\overline{Q_7}$	20 V_{CC}	Measured Terminal All outputs
7	Functional	3014	90.1	B	X	B	A	X	X	B	A	X	4/	
T_C = +25°c	3/		90.2	A	H	B	A	L	H	B	A	L	"	"
			90.3	B	H	B	A	L	H	B	A	L	"	"
			90.4	A	H	B	A	L	H	B	A	L	"	"
			90.5	B	H	B	A	L	H	B	A	L	"	"
			90.6	A	H	B	A	L	H	B	A	L	"	"
			90.7	B	H	B	A	L	H	B	A	L	"	"
			90.8	A	H	B	A	L	H	B	A	L	"	"
			90.9	B	H	A	A	L	H	B	A	L	"	"
			90.10	A	L	A	A	L	H	B	A	L	"	"
			90.11	B	L	A	B	L	H	B	A	L	"	"
			90.12	A	L	A	B	H	H	B	A	L	"	"
			90.13	B	L	A	B	H	H	A	A	L	"	"
			90.14	A	L	A	B	H	L	A	A	L	"	"
			90.15	B	L	A	B	H	L	A	B	L	"	"
			90.16	A	L	A	B	H	L	A	B	H	"	"
			90.17	B	L	A	B	H	L	A	B	H	"	"
			90.18	A	L	A	B	H	L	A	B	H	"	"
			90.19	B	L	A	B	H	L	A	B	H	"	"
			90.20	A	L	A	B	H	L	A	B	H	"	"
			90.21	B	L	A	B	H	L	A	B	H	"	"
			90.22	A	L	A	B	H	L	A	B	H	"	"
			90.23	B	L	A	B	H	L	A	B	H	"	"
			90.24	A	L	A	B	H	L	A	B	H	"	"
			90.25	B	L	B	B	H	L	A	B	H	"	"
			90.26	A	H	B	B	H	L	A	B	H	"	"
			90.27	B	H	B	A	H	L	A	B	H	"	"
			90.28	A	H	B	A	L	L	A	B	H	"	"
			90.29	B	H	B	A	L	L	B	B	H	"	"
			90.30	A	H	B	A	L	H	B	B	H	"	"
			90.31	B	H	B	A	L	H	B	A	H	"	"
			90.32	A	H	B	A	L	H	B	A	L	"	"
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.													

See footnotes at end of device type 06.

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